FYP Progress Meeting

Supervisor: Prof. AN Liang
Reporter: YIP Tsz Yeung Oscar, SEN Bijoy, ZHAO Xuecen
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THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Outline

1. Personal progress
   - Jack(ME)
   - Oscar(BME)
   - Bijoy(EEE)
2. Team collaboration (Jack)
3. Purchase list (Bijoy)
4. Project timeline (Oscar)
5. Questions

2. Personal Task (Jack)

> **Design the passive fuel cell stack**

> **Calculate the suitable size and reaction area of the fuel cell.**

> **Fabricate fuel cell components.**

> **Integrate the fabricated components into a functional stack.**

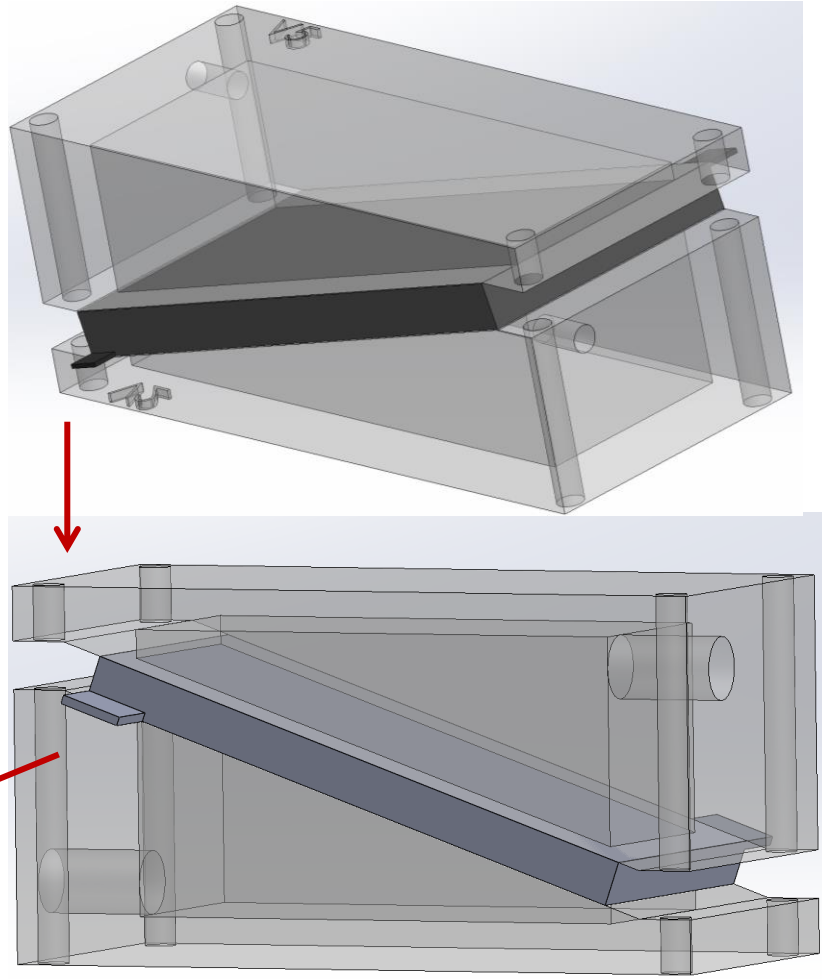
> **Assembly of the toy car.**

> **Train the AI model in collaboration with our EEE teammate.**

*Part highlighted are the task that I was working on during these two weeks

3D printed prototype one

Design from last meeting:



- Split the parts for easier 3D printing
- Equation added to the dimension for easier size modification
- Larger fuel hole for easier refueling

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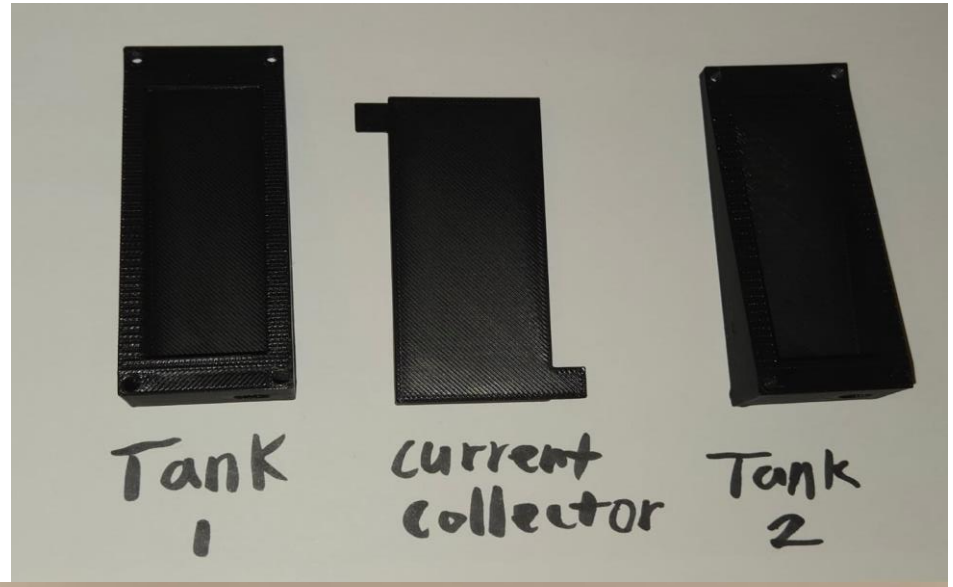
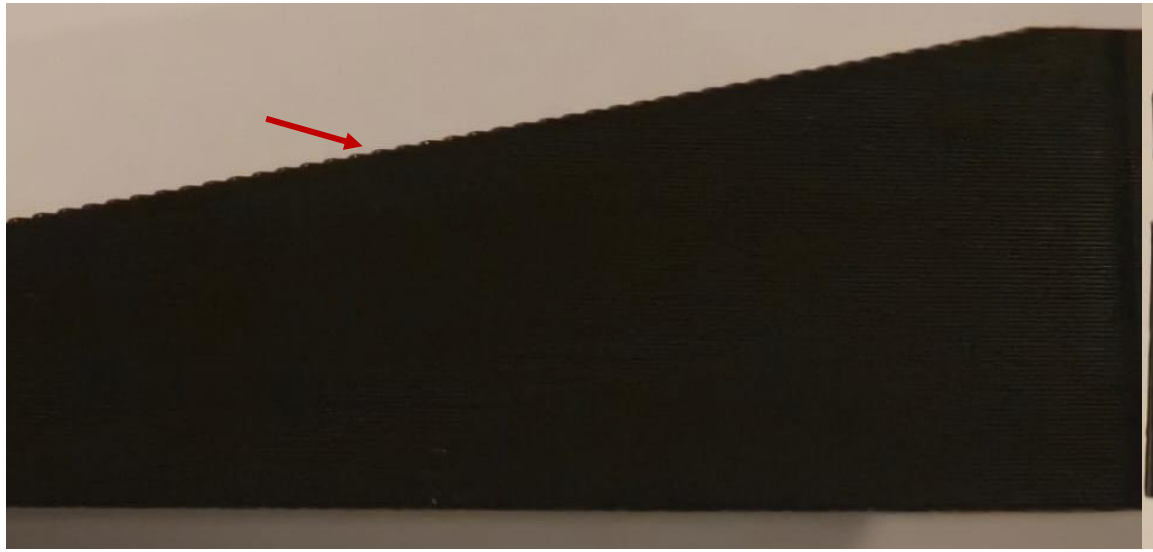
3D printed prototype one

> Printed with IC maker's lab Bambu printer

> There were a few problems:
  - Uneven edge on the tank
  - Current collector position not stable
  - Screw hole too small for conventional M3 screws

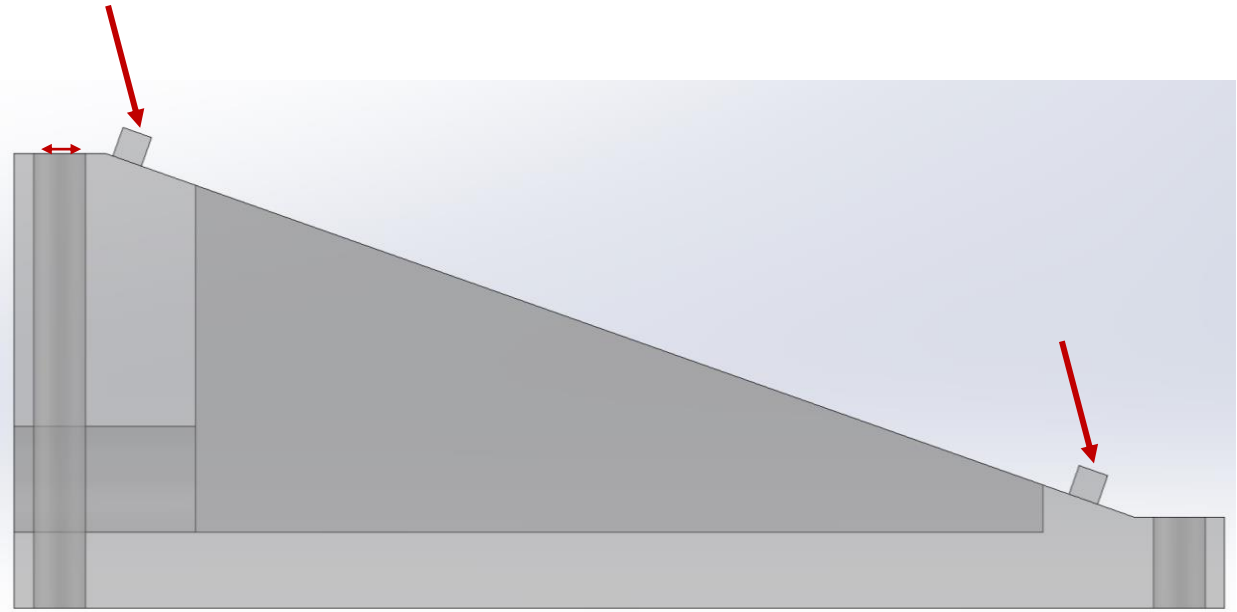
> The design will have problem with leakage, no testing needed





Evaluate and generate second prototype

- Enlarge the fuel tank size
- Enlarge the screw hole
- Extra support for the current collector
- Change the print orientation of the design to avoid rough edge



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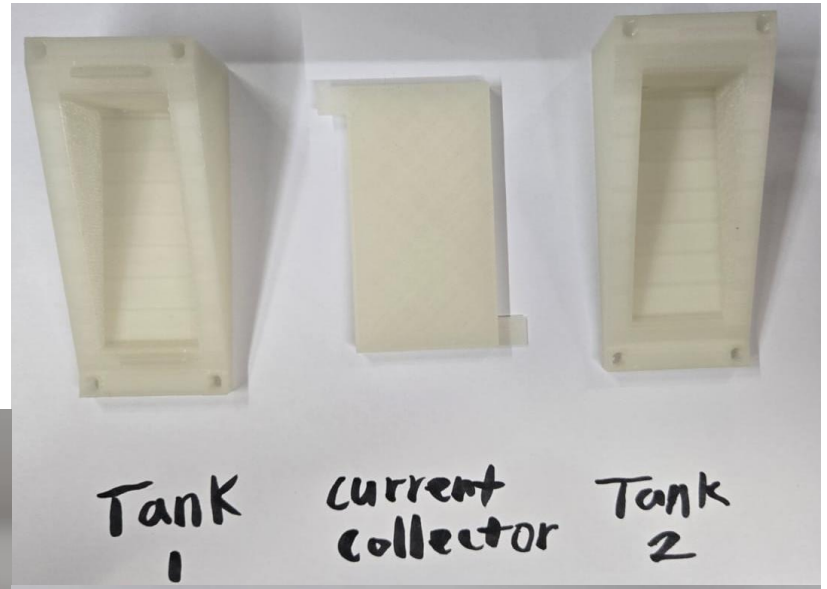
3D printed prototype two

> Most of the problem from 1st prototype is fixed in this design
  - ✓ Smooth edge in contact with current collector
  - ✓ Larger screw hole that fits a M3 screw
  - ✓ Fix the current collector in place when assembled

> Testing

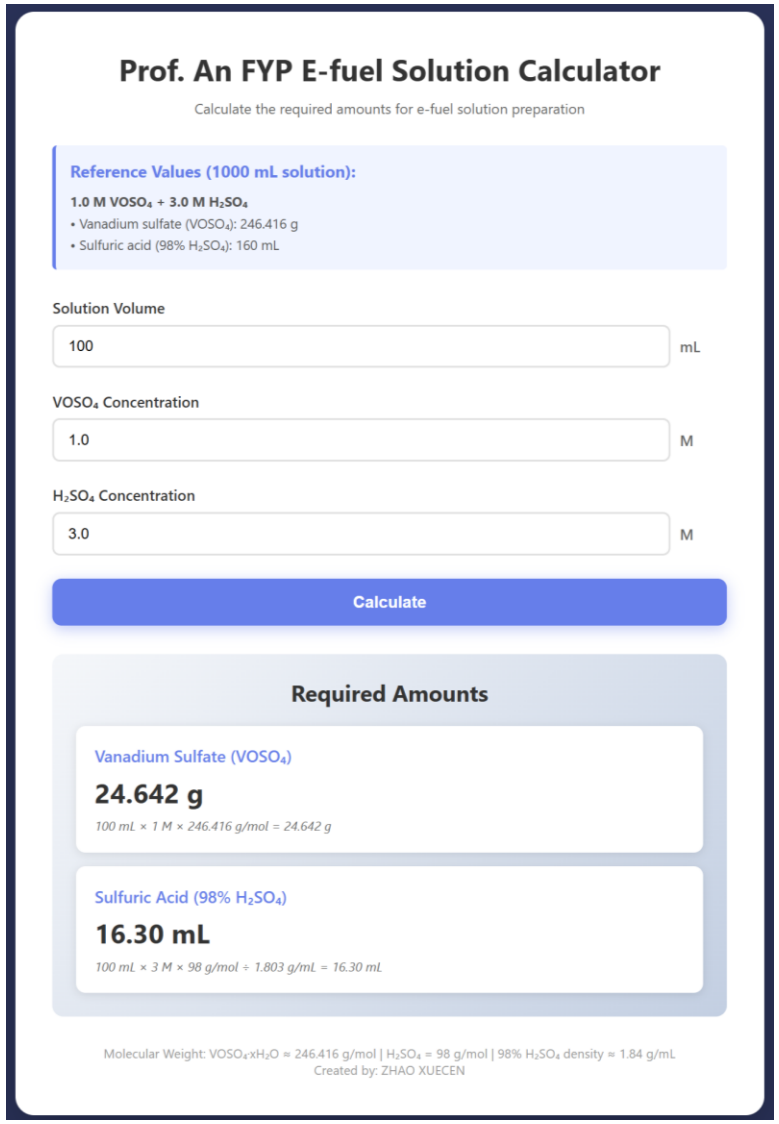


E-fuel calculation website

> https://zhao255.github.io/ZHAOpersonal-website/e-fuelcal.html

> I made this website because calculating concertation is annoying.

> It calculate the require amount of Vanadium and sulfuric acid needed from concentration.



2. Personal Task (Oscar)

Analysis the e-fuel performance

Testing we done:

1. 1 M Vanadyl sulfate solution with 4 M H_2SO_4

Retest we done:

1. 1 M Vanadyl sulfate solution with 1 M H_2SO_4
2. 1 M Vanadyl sulfate solution with 3 M H_2SO_4

2. Analysis in October

Analysis the e-fuel performance

We discover:

1 M sulphuric acid have the best performance

Which is not logical → Retest the performance

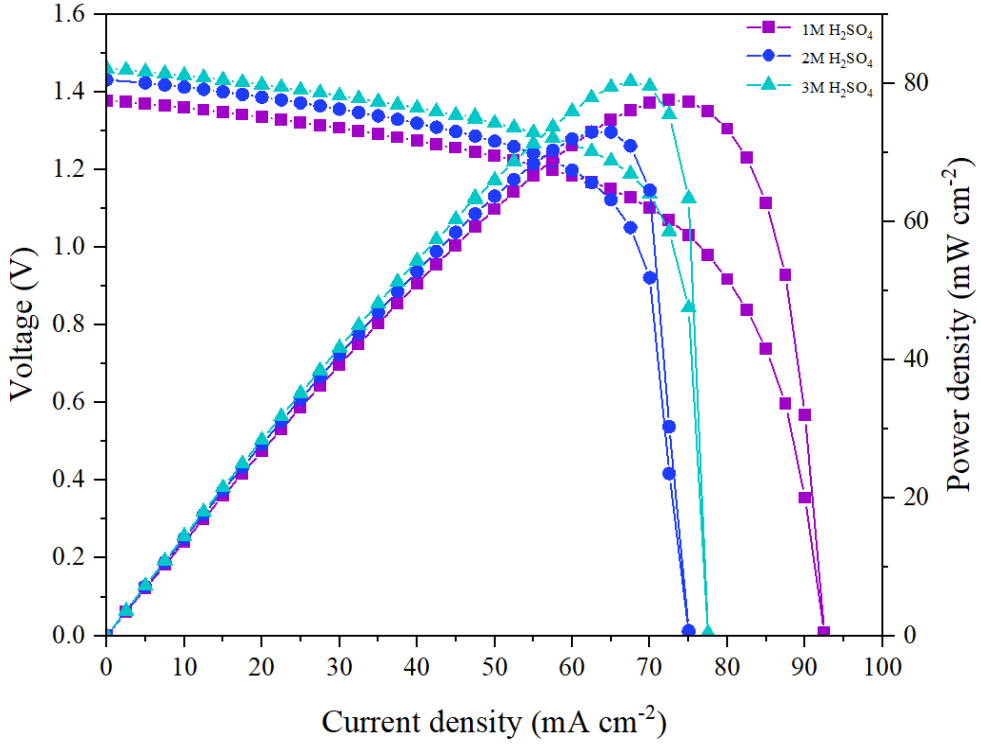


Fig.1 Effect of sulfuric acid concentrations on cell performance analysis in 2 Oct

2. Analysis in October

Analysis the e-fuel performance

We discover:

1 M sulphuric acid have the best performance

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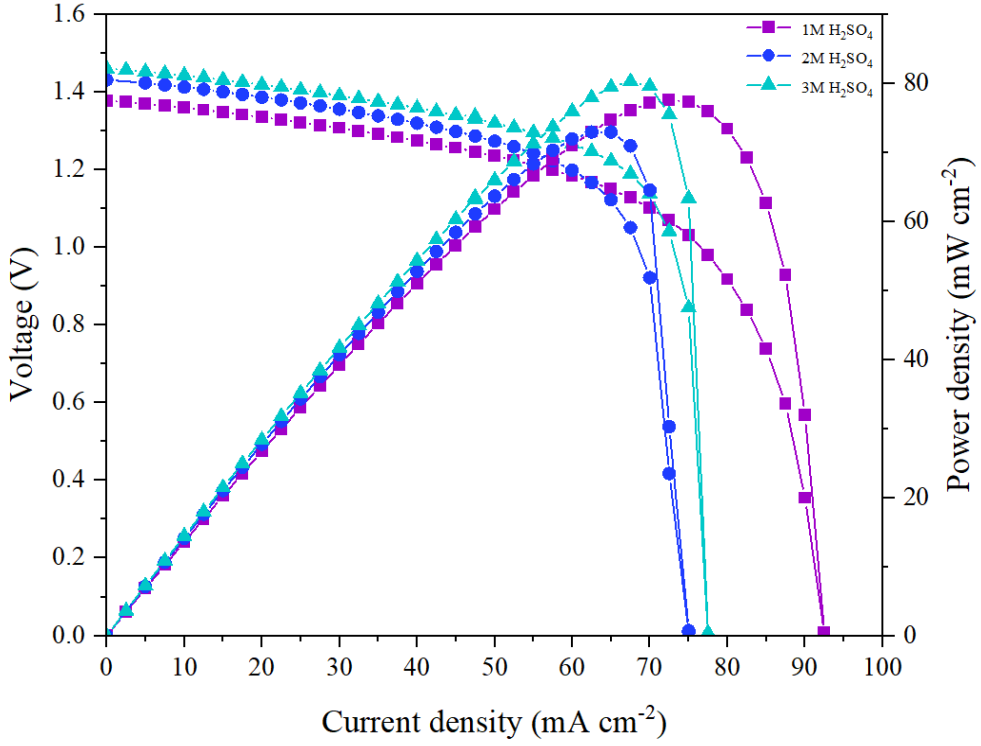


Fig.1 Effect of sulfuric acid concentrations on cell performance analysis in 2 Oct

2. Analysis in October

Analysis the e-fuel performance

After all retest We discover:

1. 1 M sulphuric acid still have good performance
2. 3 M and 4 M are better
3. 4 M also show the highest power

The result become logical, and 2 M can retest again

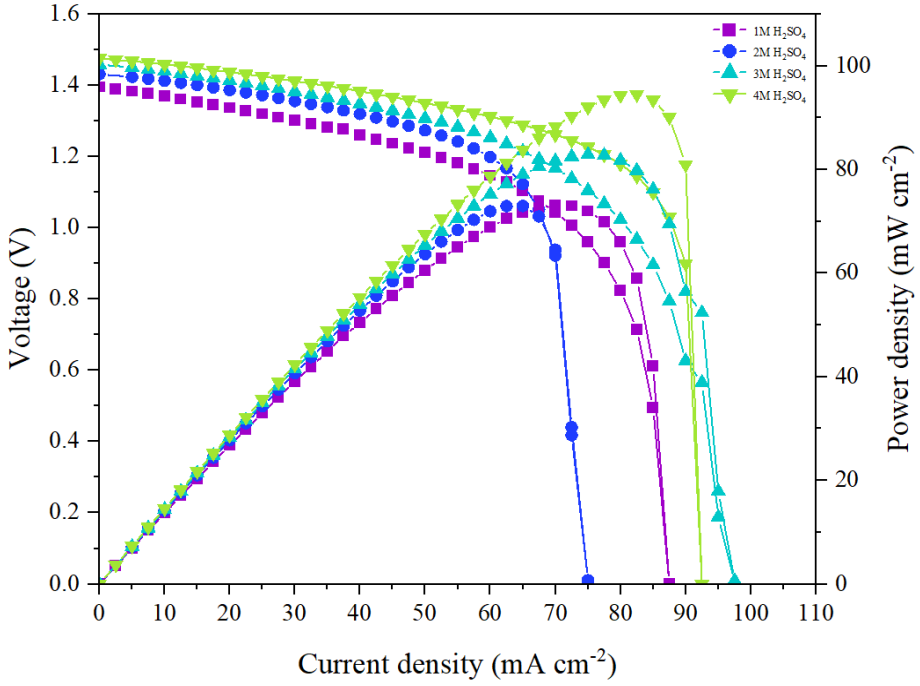
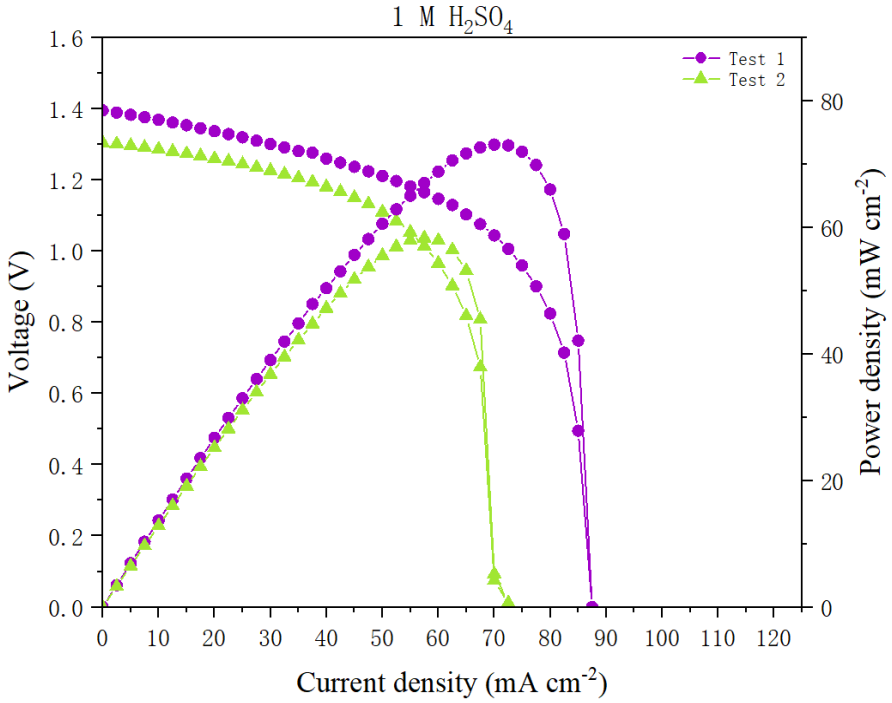


Fig.2 Effect of sulfuric acid concentrations on cell performance analysis in 9 Oct

2. Analysis in October

(a)



(b)

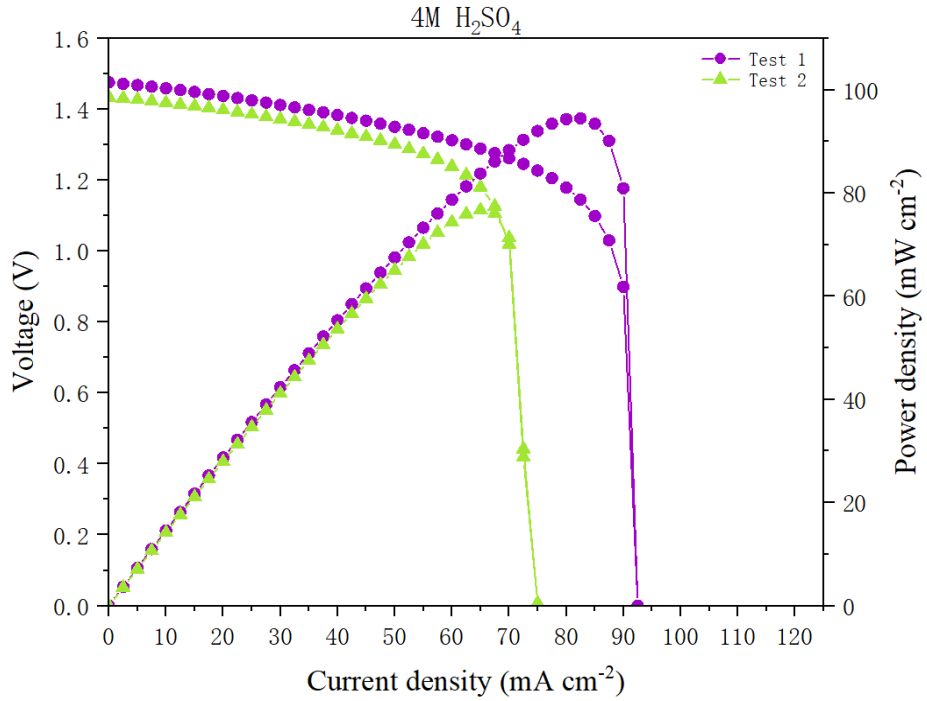


Fig.3 Decay curve of **a** 1 M sulphuric acid **b** 4 M sulphuric acid

2. Analysis in October

The second analysis was done after 10 minutes of the first test

1. The power will be lower
2. The current density and OCV lower

Which means we need to use it as fast as possible for maximum power supply.

2. Analysis in October

Continue work in the end of October

1. 2 M sulphuric acid retest
2. 5 M sulphuric acid test

In November

After we found the best concentration of sulphuric acid, we can change other parameter such as concentration of vanadyl sulfate solution

2. Personal Task (Bijoy)

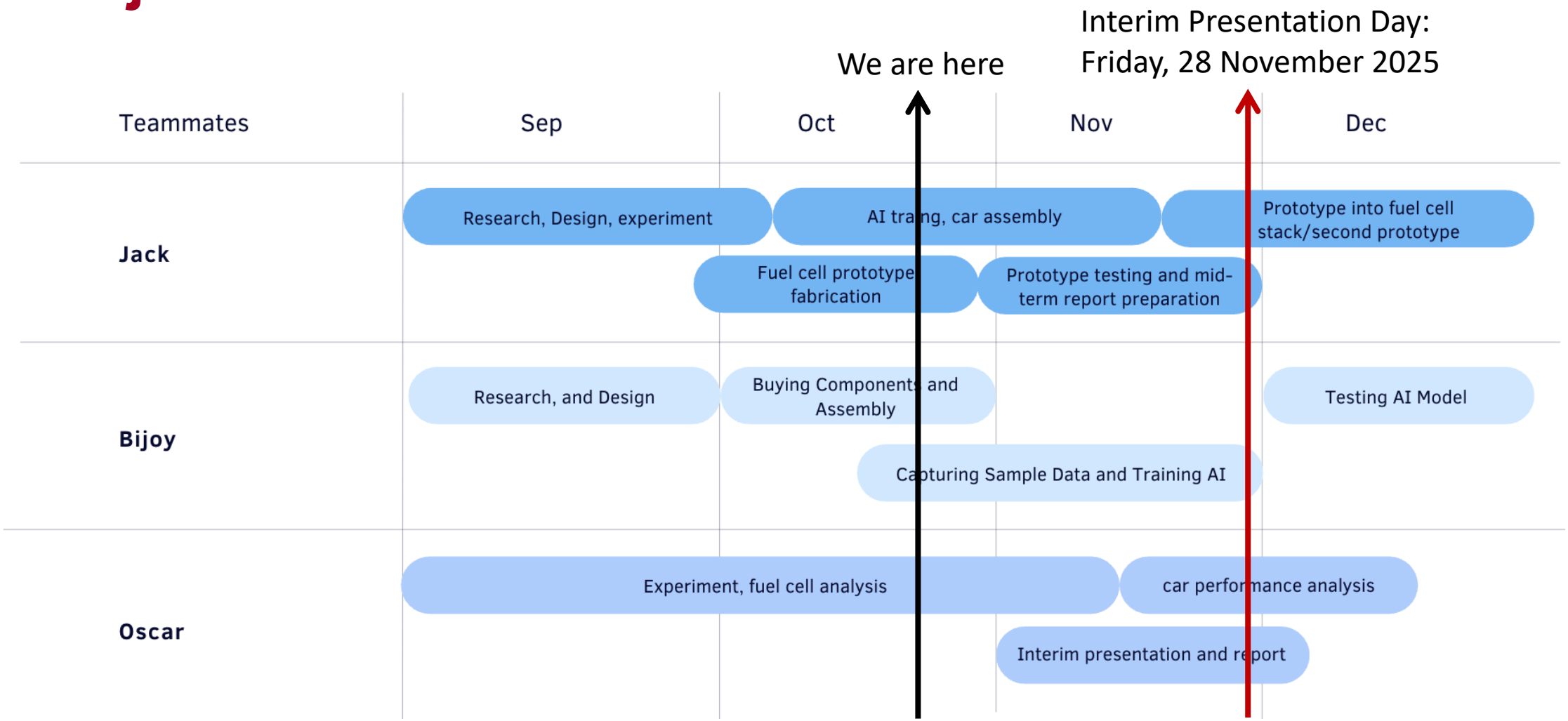
- **Assemble the overall car**
- **Start collecting sample data**
- **Begin building the AI model**

- Highlighted parts are in progress

3. Team collaboration

- **Working closely with the international team**: Since we share the same set of equipment and are doing similar experiment, we share our lab experience and compare our data for better accuracy.
- **Have regular online and offline meeting**: We discuss our progress and share our opinion on the project with meetings.
- **Always use WhatsApp for effective communication**: When we encounter problems and issues, we always contact each other to seek solutions and arrange meetings if necessary.

4. Project Timeline Chart



YIP Tsz Yeung 22077083D

Thanks for your listening !